Improving Forgery Traces via Guided Spectral Feature Reconstruction — Enhanced Forensic Trace Generation

Wu et al. (2026), in their paper *Robust Speech Forgery Detection via Enhanced Forensic Trace Generation*, present a framework for improving voice forgery detection by means of spectral feature reconstruction and amplifying the information that distinguishes real from fake speech. The study addresses the limitations of approaches based on pre-computed acoustic features, such as LFCC, CQCC and Log Power Spectrum (LPS), which offer computational efficiency and stability but may struggle to generalize to recording conditions and forgery attacks not included in training. At the same time, the researchers note that deep models may improve the ability to represent information, but sometimes require significant computational resources and show a decline in performance under distribution shifts. Accordingly, the study proposes improving the quality of the features fed into the classifier by means of a dedicated reconstruction process, instead of relying on the original features alone.

The proposed method is based on a Vector Quantized Variational Autoencoder (VQ-VAE), used to reconstruct spectral features from a discrete latent representation. The central assumption is that the reconstruction process can highlight inconsistencies and acoustic traces related to synthesized speech, thereby making their detection easier for a classifier. In order to reduce the risk of information loss during the reconstruction, the researchers combine the original features with the reconstructed features. In the experiments, the combination is carried out with an equal weight of 0.5 for each of the two information sources, and the resulting representation is passed to a ResNet-based classifier.

The central methodological novelty is the **Knowledge-Driven Generation Optimization** strategy, in which the Encoder of a pre-trained Anti-Spoofing classifier is used to guide the training process of the VQ-VAE. The Encoder's weights remain frozen, and the representations it produces serve as supervisory signals for the reconstruction model. While the VQ-VAE reconstructs spectrograms of real speech, examples of fake speech are incorporated into the supervision process to encourage discriminative separation in the feature space. For this purpose the researchers propose three complementary Loss components: **True Speech Consistency Loss**, which encourages similarity between the real speech representation and its reconstruction; **Triplet Loss**, which encourages separation between real and fake speech representations relative to the reconstructed representation; and **Positive Sample Constraint**, which limits the distance between the real speech representation and the representation of its reconstruction.

The researchers evaluated the method by training on ASVspoof 2019 Logical Access and testing on the evaluation sets of ASVspoof 2019 LA and ASVspoof 2021 LA. In the experiments, three feature families were examined — LFCC, CQCC and LPS — both in their original form and after reconstruction and combination by means of VQ-VAE. On ASVspoof 2021 LA, combining the reconstructed features improved the EER in all three families: for LFCC a decrease from 26.89% to 20.87% was obtained, for CQCC from 23.29% to 19.33%, and for LPS from 22.64% to 14.04%. The last improvement, of 8.60 percentage points, represents a relative decrease of about 38% in EER compared with a ResNet using original LPS features alone. Under these evaluation conditions, FVQ-LPS also achieved a lower EER than the comparison models appearing in the paper, including AASIST, which achieved 15.84%.

Ablation experiments supported the contribution of the proposed training strategy. Removing the Triplet Loss component increased the EER on ASVspoof 2021 LA from 14.04% to 20.92%, while removing the True Speech Consistency Loss increased it to 16.00%, and removing the Positive Sample Constraint increased it to 15.30%. These results indicate the special importance of the class-separation component. In addition, a comparison of Autoencoder, Variational Autoencoder and VQ-VAE reconstruction models showed that VQ-VAE achieved the best performance under the experimental conditions examined. Cross-Feature experiments also showed that combining the original and reconstructed features can be more beneficial than using either of them separately.

Despite the positive results, the method has several limitations. First, the improvement in generalization was demonstrated within a transition from ASVspoof 2019 LA to ASVspoof 2021 LA, and was not tested directly on ASVspoof 5 or on forgery datasets based on Neural Audio Codecs such as CodecFake. Second, the reconstruction process adds computational complexity: the proposed system includes about 5.67 million parameters and 3.12 billion MAC operations, compared with about 2.34 million parameters and 1.25 billion MAC operations in the base ResNet. Accordingly, the improvement in detection ability comes with an additional computational cost. Likewise, the results do not prove that reconstruction by means of VQ-VAE will help similarly for every acoustic representation, detection architecture or dataset.

For the present study, the central contribution of the paper is in presenting a complementary approach for improving existing acoustic features by means of task-guided reconstruction. One can examine whether combining original and reconstructed features improves forgery detection and the generalization ability between ASVspoof 5 and CodecFake, while comparing to approaches based on MFCC, CQCC, AASIST and pre-trained SSL models such as wav2vec 2.0 and WavLM. However, given the limited size of the course dataset and the training complexity of VQ-VAE, the method is, at this stage, suited to a possible extension experiment after establishing a stable Baseline and evaluating the pre-trained models. This gradual approach will make it possible to examine whether the possible improvement in EER justifies the additional resources and implementation complexity.

Bibliographic source:

Wu, S., Cheng, Y., Wang, R., & Yan, D. (2026). *Robust Speech Forgery Detection via Enhanced Forensic Trace Generation*. Signal, Image and Video Processing, 20, Article 130. https://doi.org/10.1007/s11760-026-05167-y